Diagnosis-Guided Compressed Latent Replay for Continual Document Understanding

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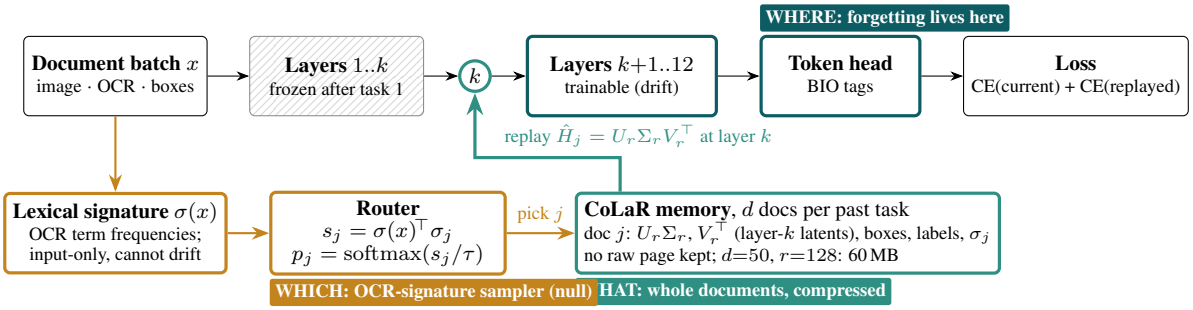


Figure 1. **Diagnosis-guided compressed latent replay (CoLaR).** The anatomy of Sec. 4 fixes *where* to act (head and late layers) and *what* memory must hold (whole documents, Sec. 7): each past document is stored as a truncated SVD of its layer-k activations (60 MB per d=50 task) and rebuilt during replay. A drift-free OCR-signature sampler probes *which* stored document to replay and is reported as a three-seed null (Sec. 7).

Abstract

Continual learning for visually-rich document understanding is dominated by raw-exemplar replay, which deployed pipelines cannot keep for privacy reasons, and by buffer-free remedies ported from unimodal classification that treat the network as a homogeneous block. We take a diagnosis-first route. Per-group Fisher displacement and layer-wise CKA across four backbones (LayoutLMv3, LiLT, BROS, BERT), three scenarios, and three seeds show that forgetting in continual key-information extraction is output-localised*—concentrated at the shared classifier head and late encoder layers—and* architecture-general*: the naive-to-joint gap agrees to within* 2.5 *F1 points across the four. Protecting the locus does not remove forgetting; it* relocates *the drift to whatever stays plastic, so a buffer-free fix at the head alone cannot work and five such families we built or ported each fail for the same reason. Guided by the diagnosis, we propose* ***CoLaR****: a replay memory that banks each past document's hidden activations at an early frozen boundary (layer k) and stores them as a truncated SVD $U_r\Sigma_r V_r^\top$ computed per document. The intact (feature, position, label) binding of a document is the ingredient every marginal summary destroys (we quantify it at* +27 *AA), and per-document rank-r keeps that binding at a fraction of the storage. CoLaR retains* 87.6 *AA on domain-incremental DIL at* 60 *MB, matching raw-exemplar replay (*87.9*) and raw latent replay (*87.3 *at* 163 *MB) within* 1.1 *of the joint oracle (*88.7*), stores no raw document bytes, and inherits the diagnosis's* 90–160× *Fisher-displacement reduction at the head. A drift-free OCR-signature router over the same memory was pre-registered as a sampler improvement and does not survive three seeds (mean* +0.5 *pp, inside the uniform spread), so the memory-only variant is the headline. Code and runs will be released on acceptance.*

1. Introduction

Organisations process a continuous stream of visually-rich documents—forms, invoices, receipts, contracts—whose meaning depends not only on the words but on where those words sit on the page. Key-information extraction (KIE) from such documents is cast as token-level sequence labelling over a tri-modal input (OCR text, 2-D layout, page image), and multimodal document encoders such as LayoutLMv3 [12], LiLT [35] and BROS [11] define the state

of the art. Deployed systems are not static: new document types appear, schemas are extended, and the input distribution drifts, while retraining on all history is blocked by cost and by privacy constraints on the earlier data. This is continual learning (CL), and its central obstacle is catastrophic forgetting [23].

The CL literature offers a mature toolbox—importance-weighted regularisation [13], distillation from past models [18], exemplar replay [3, 30], task prompts [32, 38, 39] and per-task low-rank adapters [37]—but almost all of it was developed on unimodal image or text classification, and it treats the network as one homogeneous block: a single quadratic penalty across all parameters, whole-network output matching, or a depth-wise freezing cut chosen by hand. A tri-modal document encoder is manifestly *not* homogeneous. It contains a text stream, a visual stream, layout embeddings, shared attention blocks at increasing depth, and a task head, trained on different signals with different inductive biases. The modern parameter-efficient and modular families do not resolve this; they sharpen it, because deciding *which* weights to adapt or *where* to inject a prompt presupposes knowing which part of the encoder is at risk.

Problem and approach. Under continual token-level KIE—a domain-incremental sequence over FUNSD, SROIE and CORD, a class-incremental split of CORD, and a mixed sequence that alternates the two—privacy constraints on earlier documents rule out raw-exemplar replay. The literature's buffer-free alternatives perform below replay by a large margin, and no existing analysis says why in a document setting. We follow a diagnosis-first route. First we localise forgetting inside a document encoder (Sec. 4), then show that the diagnosis explains the remedy ordering, including a backbone-dependent sign flip in parameter-space remedies (Sec. 5). The diagnosis makes a sharp prediction: because the head is where forgetting lives *and* the drift relocates when the head is protected, a buffer-free fix cannot work; we falsify five consolidation families against this prediction (Sec. 6). Finally we read off what a buffer *must* contain, and build it (Sec. 7): per-document SVD-compressed latent replay (**CoLaR**) preserves the (feature, position, label) binding of a document while compressing it, matches raw-exemplar replay at $2.7\times$ less storage, and never leaves raw document bytes in the memory.

Contributions.

1. A multi-backbone, multi-scenario *anatomy* of forgetting in document IE: output-localised, architecture-general, and *conserved* under architectural constraint—protecting the diagnosed locus relocates the drift to whatever remains plastic (Sec. 4 and Fig. 3).
2. A prediction-tested benchmark of six classical strategies over four backbones and three scenarios ($68/72$ cells, three seeds), with the diagnosis explaining both the global ordering (replay reaches the joint bound; buffer-free methods stay at the naive floor) and a 17-point sign flip in EWC's benefit across backbones (Sec. 5).
3. A convergent negative result over five buffer-free families with a single mechanistic cause—a readout-marginal snap on a class-asymmetric substrate—closed by registered kill-tests (Sec. 6).
4. **CoLaR**: per-document truncated-SVD latent replay guided by the diagnosis. It stores no raw bytes, reaches 87.6 AA at 60 MB on DIL (87.8 over three seeds, within 1.1 of the joint oracle 88.7 and matching raw-exemplar ER at $2.7\times$ less storage), and quantifies whole-document consistency as the necessary buffer ingredient ($+27$ AA single-variable control). A drift-free OCR-signature sampler is pre-registered and reported as a three-seed null (Sec. 7).

2. Related Work

Continual-learning families. Following Wang et al. [36], remedies fall into regularisation (EWC [13], SI [42], MAS [1]; output-side distillation, LwF [18]), replay (GEM [20], ER [30], DER++ [3], iCaRL [29]), optimisation-geometry (GPM [31], OGD [5], C-Flat [2]), representation-based methods over a frozen backbone (L2P [39], DualPrompt [38], CODA-Prompt [32], O-LoRA [37], CL-LoRA [10], RanPAC [24]), and architecture-based isolation [21, 41]. Replay is empirically the strongest family [22, 36] but stores raw data, which document deployments cannot keep. The buffer-free alternatives we test in Sec. 6 are drawn from the strongest current members of each line: model merging with classifier realignment (LCA [33]), subspace transfer, and exemplar-free *feature* replay—class-conditional Gaussians in a frozen feature space (PASS, FeTrIL, FeCAM [7, 27, 49]).

Latent and compressed replay. CoLaR descends from a line that stores mid-network activations instead of raw inputs. REMIND [9] freezes the lower stack and quantises later-layer features with product quantisation. Latent Replay [26] banks activations at a chosen layer and injects them during training, in CORe50-style streams. Brain-inspired generative replay [34] replaces exemplars with a VAE sampled per class. Each stores *one summary per token* and is trained for class labels; CoLaR stores *one SVD per document* and is trained for token-level BIO, so it must keep the binding between a token's feature, its position and its label—which the diagnosis of Sec. 7 shows is exactly what every marginal summary destroys.

Where forgetting lives. Ramasesh et al. [28] localise forgetting in split image-classification tasks by resetting or

freezing blocks of layers and find it concentrated in the deeper layers nearest the output; Zhai et al. [43] show the same picture carries over to multimodal LLMs used as classifiers. Multimodal CL benchmarks add hints that architecture shapes forgetting—dual-stream models forget less than single-stream ones and deeper layers forget more [45]—and per-layer budget routing (D-MoLE [6]) implicitly assumes non-uniform need without measuring it. None of these attributes forgetting to the components of a single tri-modal encoder, tests whether the location is architecture-invariant, or asks what happens to the drift when the located component is protected. We take the checkpoint-pair, per-layer stance of Ramasesh et al. [28] and extend it along two axes—component and backbone—and add the protection/migration experiment.

Continual information extraction. Continual NER confronts the semantic shift of the "O" tag with distillation and pseudo-labelling (CPFD [44], CFNER [47], ICE [19], SpanKL [48]) or by generating spans (GenCNER [40]); continual RE pairs episodic memory with relation prototypes (EMAR [8], RP-CRE [4], CRL [46]) or uses prompt pools under task-incremental assumptions (WAVE-CRE [16]). All are sentence-level and text-only. Document-level CL is nascent: GDI-Bench [17] shows an MLLM forgets under fine-tuning and mitigates it by *heuristic* layer freezing; continual table detection [25] uses plain ER; ProtoNER [15] reaches LayoutLM-family encoders only in a few-shot regime. No prior work diagnoses which component of a document encoder forgets, and none evaluates the classical CL families on multimodal KIE under one protocol across backbones.

3. Setup

Task. A document is a triple $D = (W, B, I)$ of OCR tokens, normalised bounding boxes and the page image; the model f_θ emits BIO labels per token and is trained with token-level cross-entropy. Performance is entity-level F1 with BIO-aware span matching (`seqeval`). In the continual setting the learner sees tasks $\mathcal{T}_1, \ldots, \mathcal{T}_T$ and at task t may access only $\mathcal{D}_t$ and, for replay methods, a bounded memory. OCR and boxes are dataset-provided throughout, so no OCR engine enters the loop.

Datasets and scenarios. We use FUNSD (149/50 train/test forms, 7 BIO tags), CORD (800/100/100 receipts, 30 fine-grained classes, 61 tags) and SROIE (626/347 receipts, 9 tags). **DIL**: FUNSD→SROIE→CORD under a fixed unified schema {HEADER, KEY, VALUE, OTHER}; the label space is constant and only the input distribution shifts, so forgetting is pure representation drift. **CIL-CORD**: CORD's 30 classes introduced six at a time over five sessions in a fixed easy-to-hard curriculum; the head grows, and tokens of not-yet-introduced classes are absorbed into "O", whose meaning therefore shifts. **Mixed**: a six-session sequence alternating new-class transitions within CORD with new-domain transitions to FUNSD and SROIE, ending in a FUNSD revisit. Session details and the DIL mapping are in Appendix A.

Backbones. LayoutLMv3-base [12] is the primary tri-modal encoder (text, layout, image; single-stream). LiLT-base [35] and BROS-base [11] are layout-aware but vision-free; BERT-base is a text-only external comparator. Every method runs unchanged on all four through a shared wrapper contract (head expansion, parameter groups, gradient checkpointing); heads are plain linear layers.

Methods. The generality grid covers the six classical strategies—naive sequential fine-tuning (lower bound), joint multi-task training on the pooled data (oracle upper bound), EWC, LwF, ER and DER++—on every backbone and scenario. Defaults: EWC λ=10^3 with Fisher over 200 samples; LwF T=2; ER/DER++ buffer of 200 documents, DER++ α=β=0.5. The falsified buffer-free families of §6 and the compressed-replay controls of §7 run on LayoutLMv3/DIL unless stated. Prompt- and LoRA-based baselines (CODA-Prompt, O-LoRA, CL-LoRA, ER+C-Flat++) are reported in Appendix B.

Protocol. AdamW, learning rate 5×10^{-5}, weight decay 0.01, gradient clipping 1.0. Each task trains *to convergence*: early stopping on the task's validation entity-F1 (patience 2, δ=0.1) under a 100-epoch ceiling that never binds, restoring the best-validation checkpoint at each boundary. FUNSD and SROIE have no validation split; for them the stopping signal is read from the evaluation set, a mild optimistic bias shared by every method. Grid results are mean ± s.d. over seeds $\{42, 7, 123\}$.

Metrics and the localisation protocol. From $R[i, j]$ (F1 on task j after training task i) we report AA = $\frac{1}{T}\sum_j R[T, j]$ and BWT = $\frac{1}{T-1}\sum_{j<T}(R[T, j]-R[j, j])$; forward transfer is uninformative under label growth (zero-shot term ≈ -85). Localisation is one backbone-agnostic protocol (released as a library call) over a checkpoint pair $(\theta^{(t-1)}, \theta^{(t)})$, a partition of the parameters into groups g (by component, or by depth read off the module names), and an old-task probe set. *Localise*: per-group old-task-Fisher-weighted displacement $D_g = \sum_{\phi\in g} F_\phi^{(t-1)}(\theta_\phi^{(t)} - \theta_\phi^{(t-1)})^2$, large exactly where parameters that mattered for the previous task were overwritten, and per-token linear CKA [14] between the checkpoints at one depth-matched probe per group. *Test*: dominant group vs. each rival (Mann–Whitney over seeds, Bonferroni) and,

across backbones, a permutation test on the L1-normalised D_g profile for a shared *location*. *Perturb*: protect the located group, retrain, re-run; the anatomy predicts the profile relocates rather than shrinks. D_g is distinct from Fisher *importance* F_g, which peaks at the head for any token classifier by construction (0.22 vs. 0.005 for the next group) and is never read as forgetting.

4. Anatomy of Forgetting

Forgetting is total within one boundary, on every backbone. Under naive sequential fine-tuning the diagonal of R is strong and the sub-diagonal collapses to 1–3 F1: LayoutLMv3 loses FUNSD from 89.1 to 3.3 after one further task, LiLT 83.9→2.1, BROS 88.9→6.6. On DIL the naive floor is $41.3/40.1/40.3/39.4$ on LayoutLMv3/LiLT/BROS/BERT against the joint oracle $88.7/83.8/87.5/77.2$; on Mixed the joint-to-naive gap is $+33.7/+31.1/+31.4/+31.6$, a 2.5-point spread (Table 1).

Finding 1: output-localised. Two instruments agree. Per-token CKA between consecutive checkpoints falls monotonically with depth: 1.00 at the text/layout/patch embeddings, 0.78 over early encoder layers, 0.34 mid, 0.25 at the last layer and 0.16 at the classifier. Old-task-Fisher-weighted displacement D_g (Fig. 2) is one to four orders of magnitude larger at the head than at any encoder bucket, on all four modality-mask conditions of LayoutLMv3, on BERT, LiLT and BROS. A per-component test rejects the null of uniform displacement with the head dominant ($p < 0.01$). The locus is not fusion-specific: the full model (C4), text+layout (C3) and genuine text-only LayoutLMv3 (C1) forget comparably (BWT -81, -88, -79), so removing the visual and layout streams barely changes the profile; forgetting is carried by the text-and-head path common to all three. The text stream is the one that supports usable optimisation at all—with text removed, two of three seeds collapse to an all-background predictor.

Finding 1′: architecture-general. Permuting LiLT's and BROS's per-depth displacement profiles against the LayoutLMv3 reference fails to reject a shared *location* ($p = 0.38, 0.51$), and BERT reproduces the same head-dominant, depth-monotone pattern. We claim only shared location: at three seeds a cross-backbone comparison of forgetting *magnitude* is underpowered, and we do not read the small differences in AA or BWT between backbones as meaningful.

Finding 2: protecting the locus relocates the drift. If forgetting lives at the head and late layers, the natural intervention is to protect them. On DIL/LayoutLMv3 (seed 42) freezing everything after task 1 gives AA 41.1, BWT -0.0: nothing is forgotten and nothing is learned. Freezing only the diagnosed head and late layers, leaving the early and mid encoder plastic, gives AA 38.8 with BWT -76.8—naive-level forgetting—while early/mid CKA, which is ≈ 1.0 under naive training, collapses to 0.19. The drift did not disappear; it migrated to whatever remained plastic. A control that adds a memory to the frozen locus without protecting it (38.9, -75.6) is indistinguishable, so drift, not the absence of a memory, is the bottleneck. This is why uniform protection fails too: a depth-scaled consolidation penalty spread equally across every bucket collapses to 42.6 AA (it cannot fit the tasks), whereas the same budget concentrated on the head/late locus recovers 84.7, and head-only (86.1) or late-only (86.2) each recover essentially all of it (three seeds). We read this as a conservation statement: under architectural constraint the total forgetting is redistributed, not removed. Finding 2 is established on DIL with LayoutLMv3 (seed 42 for the freeze arms, three seeds for the consolidation probe); Fig. 3 extends the freeze arm to three task orders and a long class-incremental stream.

Prediction. The anatomy is output-facing and conserved under constraint. It therefore predicts that (i) only a mechanism that re-grounds the head on old-task targets can recover competence, (ii) parameter-space protection—however placed—cannot, and (iii) the ordering should transfer across backbones because the locus does. The next three sections test each.

5. The Anatomy Predicts the Remedy Ordering

Table 1 reports the six classical strategies on four backbones and three scenarios, three seeds each; 68 of 72 cells are complete, and the 4 missing cells are marked rather than imputed.[1]

Replay re-grounds the head and reaches the oracle on DIL, on every backbone. ER and DER++ match the joint bound on all four backbones—$87.9/88.0$ vs. 88.7 on LayoutLMv3, $87.8/87.6$ vs. 87.5 on BROS, $77.1/77.4$ vs. 77.2 on BERT, $85.7/84.4$ vs. 83.8 on LiLT[2]—with BWT of -1 to -3.5. The independent per-dataset upper bound (three models, one per dataset) is $88.2/85.3/88.8/77.3$ on LayoutLMv3/LiLT/BROS/BERT—within 0.5 of pooled joint on LayoutLMv3, *above* it on LiLT and BROS, and tied with the replay cells—so the transfer headroom beyond the independent-deployment baseline is sub-point, and the question for a continual learner is whether a single-model solution captures it at a storage cost smaller than 3× model

[1] The missing cells (LwF on LiLT; EWC on LiLT/CIL-CORD) exceeded the memory of the 6 GB GPU that ran the secondary-backbone grid, even with mixed precision; they are reported as missing rather than imputed.

[2] Replay slightly exceeds the pooled-data oracle on LiLT; joint on LiLT appears under-fitted by the shared early-stopping rule.

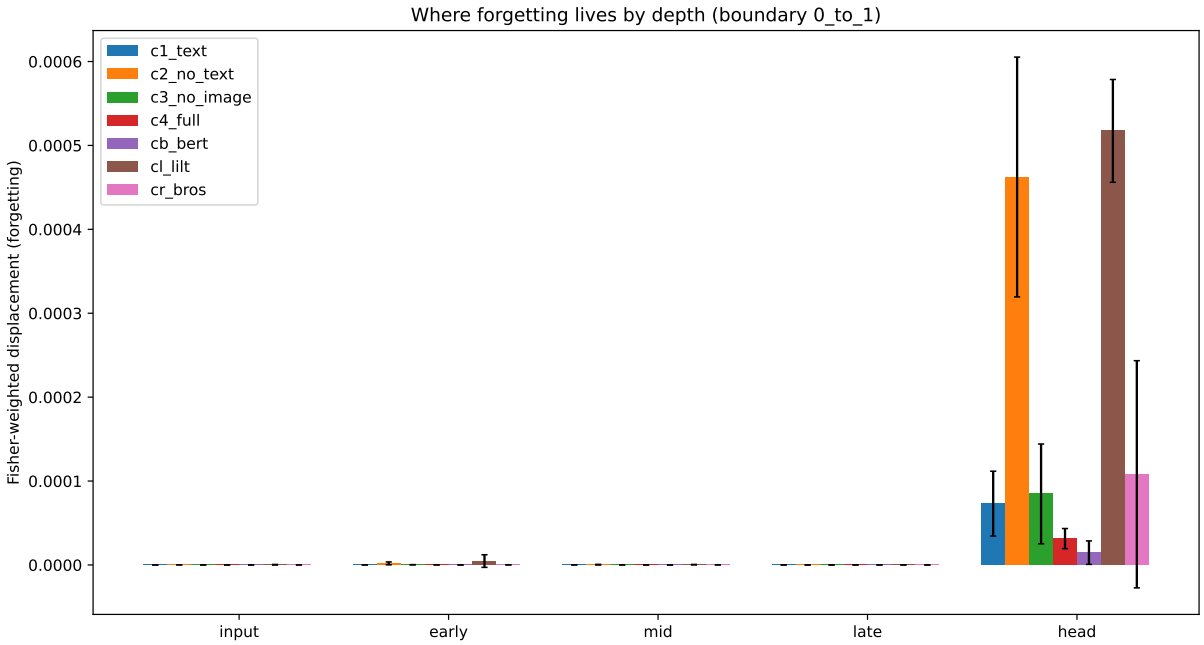


Figure 2. Where forgetting lives. Old-task-Fisher-weighted displacement D_g by depth bucket (log scale, first boundary), for the four LayoutLMv3 modality-mask conditions, text-only BERT, LiLT and BROS; mean over three seeds, ± 1 s.d. The classifier head's weighted displacement is one to four orders of magnitude above any encoder layer's on every backbone.

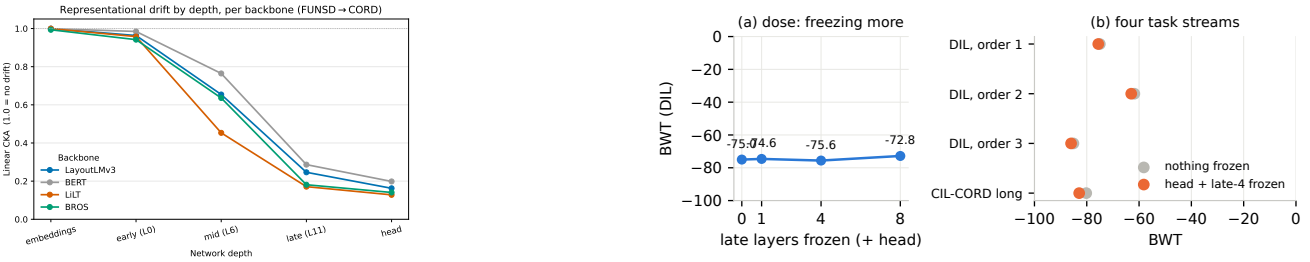


Figure 3. **Forgetting has an address, and protecting it moves it.** Left: linear CKA between checkpoints before and after the second task falls with depth on all four backbones (1.0 = no drift). Right: freezing the head and late layers leaves backward transfer unchanged, (a) whatever the number of frozen layers and (b) on three DIL task orders and a ten-session CIL-CORD stream (LayoutLMv3, seed 42, no memory). The drift relocates to the layers left plastic.

weights. On Mixed, replay closes most of the gap (60.3–63.0 vs. oracles 65–68 on layout-aware backbones; BERT 50.2/50.7 vs. 55.0); the class-incremental transitions inside the sequence are where residual forgetting sits.

Parameter-space protection does not. EWC and LwF remain at the naive floor on DIL for LayoutLMv3 (41.2, 43.7 vs. 41.3). Both mechanisms act on parameters or on whole-network outputs; neither places an old-task gradient on the head, and the anatomy says that is the only place it would matter. EWC's Fisher penalty, computed at the head-dominant locus, does reduce backward transfer (-2.8 vs. -73 for naive on LayoutLMv3) but at the cost of not fitting the new tasks—the same under-fitting mode as the uniform consolidation probe of §4.

A remedy's benefit is backbone-dependent; the phenomenon is not. Where the naive-to-joint gap agrees to 2.5 points across backbones on Mixed, EWC's benefit over naive spans a 17-point range with a sign flip: -11.8 on LayoutLMv3, -2.6 on LiLT, $+3.5$ on BROS, $+5.3$ on BERT (Fig. 4). LwF likewise helps CIL-CORD on BROS and BERT ($+4$, $+5$) and nothing on LayoutLMv3. A single-backbone comparison of buffer-free remedies can therefore land on either side of zero; a multi-backbone protocol is not optional for this class of claims.

Table 1. Generality grid: average accuracy (mean ± s.d., three seeds) of the six classical strategies on four backbones and three scenarios. *Indep.* is the per-dataset upper bound (one model trained independently on each dataset, three-seed macro over FUNSD/SROIE/CORD). *Joint* is the pooled-data oracle (one model, all three datasets). Independent is only meaningful on DIL, where the task stream is three distinct datasets; CIL-CORD is one dataset split by label and Mixed is a composed stream, so Indep. is undefined there and marked N/A. Grey cells did not fit the available GPU memory (see text). Replay reaches the joint bound on DIL on every backbone; regularisation and distillation stay near the naive floor; on CIL-CORD every method, replay included, sits near the floor.

Scenario	Backbone	Naive	Indep.	Joint	EWC	LwF	ER	DER++
DIL	LayoutLMv3	41.3±1.5	88.2±1.0	88.7±1.1	41.2±1.1	43.7±1.1	87.9±1.4	88.0±0.9
	LiLT	40.1±0.5	85.3±0.8	83.8±1.2	48.4±5.9	–	85.7±0.5	84.4±0.3
	BROS	40.3±1.8	88.8±0.1	87.5±1.3	45.0±4.9	35.2±3.1	87.8±0.4	87.6±1.1
	BERT	39.4±0.3	77.3±0.5	77.2±0.4	46.6±3.0	40.5±0.1	77.1±1.0	77.4±0.6
CIL-CORD	LayoutLMv3	19.0±0.2	N/A	33.3±0.3	0.5±0.6	18.8±0.2	16.0±1.4	18.2±0.3
	LiLT	19.1±0.3	N/A	32.4±0.4	–	–	17.9±0.5	18.0±1.0
	BROS	19.2±0.1	N/A	32.9±0.0	17.3±0.8	23.1±1.2	18.3±1.5	17.7±1.1
	BERT	19.3±0.1	N/A	32.4±0.2	17.9±0.3	24.3±0.3	17.1±1.6	19.2±1.9
Mixed	LayoutLMv3	33.9±0.3	N/A	67.5±0.8	22.0±9.4	35.2±1.2	60.3±5.3	60.8±2.7
	LiLT	30.7±0.7	N/A	61.8±0.6	28.1±3.0	–	57.0±1.4	55.1±2.9
	BROS	33.9±0.4	N/A	65.3±2.1	37.4±3.3	34.4±0.5	61.1±1.4	63.0±0.4
	BERT	23.4±0.3	N/A	55.0±0.6	28.6±0.9	29.4±4.2	50.2±0.6	50.7±1.3

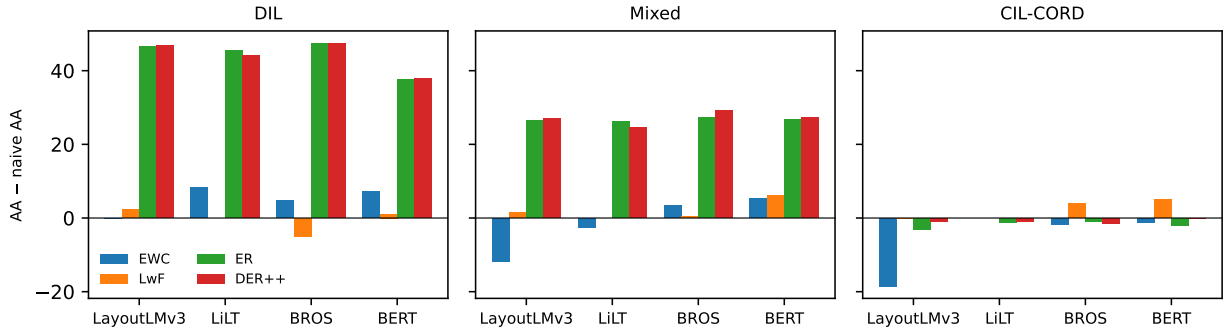


Figure 4. Remedy benefit is backbone-dependent; the phenomenon is not. Change in AA over naive for each classical remedy, per backbone and scenario (mean of three seeds). Replay's benefit is large and consistent; EWC and LwF change sign across backbones (missing bars: cells not run).

CIL-CORD is a floor regime for everyone. With six fine-grained classes per session over a single receipt domain, every method—replay included—sits at 16–24 AA against an oracle of 33, and on LayoutLMv3 EWC collapses to 0.5. Replay does not recover CIL-CORD with a 200-document buffer, and we report that rather than restrict the headline to DIL. The reason is the moving "O" boundary: what must be re-grounded is not a fixed head but a head whose background class is redefined at every session, and replayed documents carry the *old* labelling of the new classes.[3]

[3]This is the setting in which the continual-NER literature's distillation/pseudo-labelling protocols are designed to operate; we deliberately do not engineer the O-shift away.

6. Falsification: Five Buffer-Free Families

Sections 4–5 say replay works because it re-exercises the head on old-task targets. The strongest way to test that reading is to try to get the same effect without a buffer. We built or ported five buffer-free families spanning the mechanisms the literature offers, ran each on DIL/LayoutLMv3, and root-caused each failure (Table 2). None is a contribution; each is a witness.

One mechanism. A per-boundary root-cause analysis localises every failure to the readout. The head's output marginal on old tasks snaps to the just-trained task's gold

Table 2. Falsification ledger, DIL/LayoutLMv3 (naive floor 41.3, replay ≈ 88). Each family is the strongest instance we could build of its mechanism; the failure reason is from a per-boundary root-cause analysis, not inferred from the score. †run at a frozen-backbone, short-schedule (3–5 epoch) operating point whose own naive floor is given in the cell.

Family	Instance	Best AA	Why it fails
Weight merging	TIES/Fisher head merge†	21.6 (no-merge 21.9)	merge of head vectors cancels; near no-op
Merge + realignment	LCA [33] (ported)	41.9	backbone merge halves BWT; Gaussian realignment kills acquisition
Subspace transfer	HGT / CUBER (ported)†	39.8 (own naive 38.7); HGT ≤ 25.3	protected subspace grows; head unconstrained; α is a pure stability–plasticity dial
Parametric slots	LexSlot (no buffer)	42.2	isolated head slots cannot re-ground shared head
Input-anchored memory	Ledger†	FUNSD 21 / SROIE 23	own-vs-cumulative gap is training-free
Feature-Gaussian replay	LexMem v3b (positive control)	66.0	partial: recovers dense-label tasks only
	LexMem v5 (+relational)	63.2; mid-task 13.0	marginal summary breaks doc consistency

marginal (cosine 0.997–0.998 for naive and LwF, vs. 0.60 to the old task's own); replay methods stay at 0.99–1.00 to their own. Classes never re-exercised (KEY, HEADER) are extinguished to exactly 0.0 F1 under every modality mask, and the background class collapses too. Forgetting here is a *readout-marginal snap on a class-asymmetric substrate*, not representation erasure—which is why the feature-Gaussian family [7, 27, 49], exercising the head on marginal summaries, only partially recovers (Sec. 7).

Cheap corrections are registered nulls. If the snap were a mis-calibrated prior, correction should reverse it. Five kill-tests registered with explicit bars before any run—marginal matching, prior-ratio correction, per-document EM, a KL anchor to the prior-task mixture, balanced-softmax logit adjustment—recover at best −0.1 AA and leave the extinguished classes at 0.0 F1. A saturation probe explains why: old-class probability mass is $\sim 10^{-6}$, two to three orders below the branch threshold; the information is destroyed before any correction can act. Combined with the migration result of Sec. 4, buffer-free fixes at the head alone are ruled out, and the search shifts to *what minimal object a buffer must store*.

7. CoLaR: What a Document Memory Must Hold

Replay works because it re-exercises the head on inputs that keep their old joint labelling; the diagnosis of Secs. 4 and 6 says any admissible memory has to preserve that. The question is what the minimum stored object is. We answer with a falsification ladder over *latent* replay—activations banked at a frozen encoder boundary (layer k) and replayed through the plastic upper layers and head, so storage content varies while the mechanism is held fixed—and read off a memory that keeps every token's feature at its own position with its own label while compressing each document on its own (DIL/LayoutLMv3; single-seed diagnostic runs except where noted).

Whole documents retain; every summary collapses. Replaying the real banked activations of 50 whole documents per task retains AA 87.3 (k=4). Summaries of the *same* store collapse: a rank-16 spectral summary 41.9, per-class Gaussians 39.4, even real k-means centroids 36.7–39.7—so the failure is not synthesis fidelity. The single-variable control isolates the ingredient: four whole replayed documents retain 63.8 at a matched five-epoch budget, while the identical features on decoupled layout carriers retain 36.7. Whole-document binding is worth +27 AA by itself.

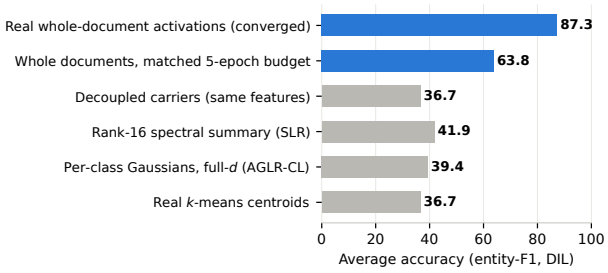


Figure 5. **What the memory must hold** (DIL, LayoutLMv3, layer k=8). Whole documents retain; summaries of the same store fall to the naive level (41.3), and the identical features on decoupled layout carriers lose 27 points against matched whole documents.

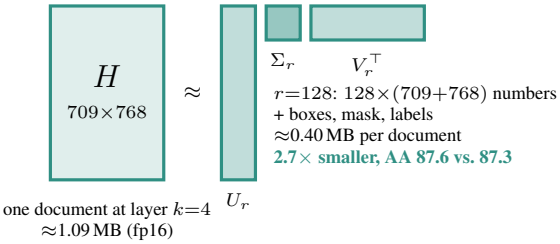


Figure 6. **CoLaR stores whole documents, compressed one at a time.** The pooled layer-k feature space is near full-rank (rank 16 keeps 43% of the variance), but a single document's activations are low-rank: its tokens share one layout and one vocabulary. Truncated SVD per document keeps every token's feature at its own position with its own label, which is the binding that every marginal summary destroys (Fig. 5).

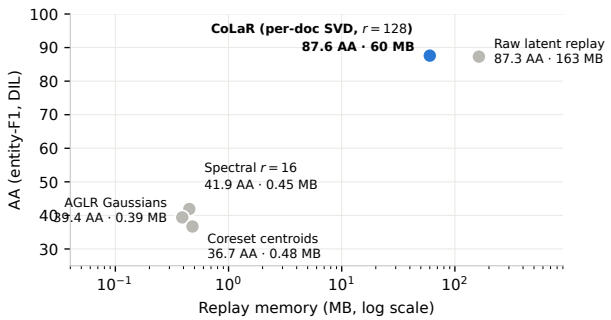


Figure 7. Replay memory versus retention (DIL, log-scale memory). Per-document SVD replay (CoLaR) matches raw latent replay at $2.7\times$ less storage because it preserves whole-document binding; every aggressive summary at $\sim$0.4 MB breaks that binding and collapses to the naive level.

CoLaR, the proposed memory. The two negatives above—every summary of a store collapses while whole documents retain—say the admissible compressor must keep each document's binding. Individual documents are low-rank even though the pooled feature space is not: one document's tokens share one layout and one vocabulary, so its layer-k activation matrix $H \in \mathbb{R}^{N\times d}$ has fast singular-value decay. We therefore **truncate the SVD per document**: $H_j \approx U_r \Sigma_r V_r^\top$, stored with the document's bounding boxes, attention mask and BIO labels. Replay rebuilds $\hat{H}_j = U_r \Sigma_r V_r^\top$ and injects it at layer k through a forward hook; the loss is CE on the current batch plus CE on the replayed documents. The memory never stores a raw image or token id. Full architecture in Figs. 1 and 6; implementation in `doccl/methods/colar.py` on top of a shared latent-replay base.

CoLaR at r=128 matches raw-exemplar replay. CoLaR retains AA 87.6 (final row $[89.2, 76.3, 97.2]$ on FUNSD/SROIE/CORD; 87.8 ± 0.5 over three seeds) at 60 MB, within 1.1 points of the joint oracle, matching raw-exemplar ER (87.9, 200 documents) at a budget that stores no private bytes, and $2.7\times$ smaller than raw latent replay at equal document count (87.3 at 163 MB); r=64 gives 80.6 at 32 MB (Fig. 7). Attempts to shift the memory-side frontier obey a conservation law—class-balanced losses 87.1, coverage-based selection 86.8, entity-weighted compression 87.1 all trade FUNSD for SROIE about 1:1 without moving AA; a three-seed check finds no lever beats the base (87.8 base = balanced 87.8 > k-centre 87.0). Read-side extensions are clean negatives (a kNN readout blend is inert at λ=0.3 and destructive at λ=1, 46.2; metaplastic consolidation is dose-dependently harmful, 85.9 $\rightarrow$ 79.5).

Routing which stored document to replay (null). The OCR vocabulary of a document is set by the input and does not drift with training, which suggests a drift-free sampler (Fig. 1): a softmax over term-frequency signature scores $s_j = \sigma(x)^\top \sigma_j$ in three directions (*near*, *far*, *task*), uniform as the zero-point. The seed-42 reading was promising—at d=50 against uniform 86.4, far 88.2 (+1.8), task 87.3 (+0.9), near 86.5 (+0.1)—so we pre-registered the single arm that cleared +1 AA (far at d=50) for confirmation. Across three seeds the gap collapses: far vs uniform at d=50 gives +1.8 (seed 42), +0.4 (seed 7), −0.6 (seed 123), mean +0.5 pp with a sign flip, inside the uniform-sampling spread. Lexical routing is reported as a null; the memory-only CoLaR number (87.8 ± 0.5) is the headline and does not depend on it.

A bounded positive on the storage side. Replaying pseudo-labelled *public* documents through the same mechanism (zero private bytes) recovers dense-label tasks (FUNSD 76.1) but not a task whose feature region the proxy pool does not cover (SROIE $\sim$ 7), topping out at 58.9: coverage-limited partial transfer. Budget-adaptive allocation of documents and rank across tasks moves along a Pareto frontier rather than beyond it (Appendix B).

8. Limitations

Scope of each claim. Finding 1 (location) rests on four backbones, three seeds, and two independent instruments; its magnitude comparison across backbones is underpowered and not claimed. The benchmark ordering (Sec. 5) is three seeds on 68 of 72 cells; the missing cells are disclosed, not imputed. Finding 2 (migration) and the buffer-content ladder (Sec. 7) are seed-42 DIL/LayoutLMv3 diagnostics with three-seed checks only at the points stated; CoLaR's headline is three-seed. Task order matters (DIL reversed: naive BWT −81 $\rightarrow$ −75); single-order results carry that caveat.

Lexical routing is a null. The pre-registered routing arm (far at d=50) does not survive three seeds (sign flip, mean +0.5 pp; Sec. 7). The headline is the memory-only CoLaR (87.8 ± 0.5) and is unaffected; what we lose is a hoped-for drift-free sampler.

What we do not claim. CoLaR is still a replay method: it needs a store proportional to the number of past tasks. CIL-CORD is a floor regime in which the O-tag shift dominates, and we have not tested whether the continual-NER distillation protocols recover it on top of CoLaR. LiLT's separate layout stream is not stored, so latent replay there is approximate.

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Diagnosis-Guided Compressed Latent Replay for Continual Document Understanding

Supplementary Material

A. Setup details

DIL unified schema. A single four-entity label space {HEADER, KEY, VALUE, OTHER} (9 BIO tags) is held fixed over FUNSD→SROIE→CORD. FUNSD: HEADER→HEADER, QUESTION→KEY, ANSWER→VALUE. SROIE: COMPANY→KEY; ADDRESS/DATE/TOTAL→VALUE. CORD: menu/sub_total/total→VALUE; void_menu/sub→OTHER. KEY originates only from FUNSD and SROIE, so the schema is VALUE-dominant; we report per-class F1 for every DIL result and treat a high aggregate as retention only if the sparse classes are retained. The reverse order is used as a robustness check.

CIL-CORD sessions. CORD's 30 fine-grained classes in five sessions of six, ordered easy-to-hard by superclass: (0) core menu fields; (1) extended menu + sub; (2) sub_total fields; (3) total / payment methods; (4) residual total + void/auxiliary. Every receipt containing at least one in-session entity is kept, with out-of-session tokens relabelled to "O", giving per-session training pools of 800/382/549/723/780 receipts. Class imbalance across sessions is deliberately left unmodified. Joint uses the full-label pool (each receipt once with all its labels) rather than the concatenation of the masked per-session copies, which would feed conflicting labels for the same document.

Mixed. Six sessions: CORD-menu → FUNSD → CORD-menu-extended → SROIE → CORD-sub_total → FUNSD-revisit; boundaries alternate new-class and new-domain transitions and the final session is an explicit retention check.

Datasets. FUNSD (`nielsr/funsd-layoutlmv3`; 149/50), CORD (`naver-clova-ix/cord-v2`; 800/100/100), SROIE (`mp-02/sroie`; 626/347). Dataset-provided OCR and boxes throughout; images 224×224; sequence length 512.

Hyperparameters. AdamW, lr 5×10^{-5} constant, weight decay 0.01, clipping 1.0, batch size 8 on the primary grid (1–2 with gradient checkpointing on the 6 GB secondary-backbone machine; replay batch 2, or 1 for DER++ on BROS, whose HF implementation lacks checkpointing). EWC λ=10^3, Fisher over 200 documents; LwF T=2, α=1; ER/DER++ buffer 200 documents, DER++ α=β=0.5. Early stopping on task val F1, patience 2, δ=0.1, cap 100 epochs (never binding; tasks converge in 4–8). Seeds 42/7/123. Primary-grid runs used bf16 autocast on a 24 GB GPU; the secondary-backbone cells ran in fp32 or fp16 on a 6 GB GPU.

Table 3. Prompt/LoRA/C-Flat++ baselines, DIL, AA and BWT, mean±std over seeds 42/7/123. Naive floor 41.3 (LayoutLMv3), ER 87.9.

	LayoutLMv3		BERT	
Method	AA	BWT	AA	BWT
L2P	30.4 ± 0.4	-11.5 ± 0.1	34.4 ± 0.7	-38.5 ± 1.0
DualPrompt	29.9 ± 0.6	-11.1 ± 0.2	34.0 ± 0.2	-40.2 ± 1.1
CODA-Prompt	32.5 ± 1.2	-10.5 ± 0.2	35.2 ± 0.7	-40.7 ± 1.3
O-LoRA	40.8 ± 0.3	-56.8 ± 1.0	37.4 ± 0.3	-53.4 ± 0.9
CL-LoRA	41.1 ± 0.3	-58.8 ± 2.1	38.6 ± 0.4	-31.6 ± 1.9
ER + C-Flat++	88.4 ± 0.1	-1.8 ± 0.6	78.3 ± 0.4	-2.2 ± 0.1

Compute. Every run fits a single GPU; the full generality grid is roughly 200 GPU-hours on a 24 GB card.

B. Additional results

Prompt-, LoRA- and flatness-based baselines. Tab. 3 reports the frozen-backbone prompt family (L2P, DualPrompt, CODA-Prompt), the orthogonal/continual LoRA family and ER with C-Flat++ sharpness-aware optimisation on DIL, three seeds, LayoutLMv3 and BERT. The prompt family forgets little but never acquires the task (AA ≈ 30–35, below the naive floor of 41.3): a frozen document encoder with a drifting shared head cannot be routed back into place. LoRA adapters acquire but forget as much as naive training does, because the adapter does not protect the head. Only the replay-based variant reaches the replay ceiling, and its sharpness term adds nothing over plain ER (88.4 vs. 87.9 on LayoutLMv3), consistent with §5: the remedy ordering is set by whether the head is re-grounded, not by the optimiser.

Per-backbone forgetting. The full BWT and AF tables for every backbone×scenario×method cell of the generality grid are generated from the released run directories (`results/table_backbone_{BWT,AF}.tex`) and will ship with the code; Table 1 already carries the AA view.

Budget-adaptive allocation is a frontier, not a gain. Allocating a fixed compressed-replay budget unevenly across tasks (documents [18, 5, 2], ranks [64, 128, 64] vs. a uniform 10 documents at rank 64; 5 epochs, k=4) gave +4.0 AA

on seed 42 while passing a pre-registered per-domain guard. On seeds 7 and 123 the same comparison gave -8.0 and $+5.7$: $+0.6 \pm 7.5$ over three seeds, a null. The seed-42 result was a point on a Pareto frontier (old-domain retention traded against the final domain at $\approx 9{:}1$), not a shift of the frontier, and we report it as such.